Guided modes in a dielectric waveguide

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1 Introduction

In this section we derive, step by step and in closed form, the two-dimensional (2D) vectorial eigenproblem for guided modes of a straight dielectric waveguide that is invariant along z. We adopt the time-harmonic phasor convention $e^{-i\omega t}$ throughout, so that $\partial_t \to -i\omega$.

2 Maxwell's Equations and Phasor Convention

In linear, isotropic media with conduction current $J = \sigma E$, the time-domain Maxwell equations are:

Gauss's Law for Electricity:

$$\nabla \cdot \mathbf{D}(\mathbf{r},t) = \rho(\mathbf{r},t) \tag{1}$$

Gauss's Law for Magnetism:

$$\nabla \cdot \mathbf{B}(\mathbf{r},t) = 0 \tag{2}$$

Faraday's Law:

$$\nabla \times \mathbf{E}(\mathbf{r},t) = -\frac{\partial \mathbf{B}(\mathbf{r},t)}{\partial t} \tag{3}$$

Ampère–Maxwell Law:

$$\nabla \times \mathbf{H}(\mathbf{r},t) = \mathbf{J}(\mathbf{r},t) + \frac{\partial \mathbf{D}(\mathbf{r},t)}{\partial t} \tag{4}$$

The **constitutive relations** for a linear, isotropic medium are:

$$\mathbf{D}(\mathbf{r},t) = \varepsilon(\mathbf{r},\omega)\,\mathbf{E}(\mathbf{r},t) \tag{5}$$

$$\mathbf{B}(\mathbf{r},t) = \mu(\mathbf{r},\omega)\,\mathbf{H}(\mathbf{r},t) \tag{6}$$

$$\mathbf{J}(\mathbf{r},t) = \sigma(\mathbf{r},\omega)\,\mathbf{E}(\mathbf{r},t) \tag{7}$$

Using the phasor convention

$$\mathbf{F}(\mathbf{r},t) = \Re\big\{\mathbf{F}(\mathbf{r},\omega)\, e^{-i\omega t}\big\} \quad \Rightarrow \quad \frac{\partial}{\partial t} \mapsto -\, i\omega, \tag{8}$$

Faraday's law (3) becomes

$$\boxed{\nabla \times \mathbf{E}(\mathbf{r},\omega) = i\omega\, \mu(\mathbf{r},\omega)\, \mathbf{H}(\mathbf{r},\omega)}. \tag{9}$$

Ampère–Maxwell (4) becomes

$$\nabla \times \mathbf{H}(\mathbf{r},\omega) = \sigma(\mathbf{r},\omega)\mathbf{E}(\mathbf{r},\omega) - i\omega\, \varepsilon(\mathbf{r},\omega)\mathbf{E}(\mathbf{r},\omega). \tag{10}$$

Defining the complex permittivity

$$\varepsilon_c(\mathbf{r},\omega) \equiv \varepsilon(\mathbf{r},\omega) + \frac{i\,\sigma(\mathbf{r},\omega)}{\omega} \tag{11}$$

we can write the frequency-domain Ampère–Maxwell equation compactly as

$$\boxed{\nabla \times \mathbf{H}(\mathbf{r},\omega) = -\, i\omega\, \varepsilon_c(\mathbf{r},\omega)\, \mathbf{E}(\mathbf{r},\omega)}. \tag{12}$$

3 Curl–Curl Wave Equations (Electric and Magnetic Forms)

From equation (9), we can express the magnetic field as

$$\mathbf{H}(\mathbf{r},\omega) = \frac{1}{i\omega}\, \mu^{-1}(\mathbf{r},\omega)\, \big(\nabla \times \mathbf{E}(\mathbf{r},\omega)\big). \tag{13}$$

Insert into (12):

$$\nabla \times \left(\frac{1}{i\omega} \mu^{-1} \nabla \times \mathbf{E}\right) = -\, i\omega\, \varepsilon_c\, \mathbf{E}. \tag{14}$$

Multiply by $i\omega$ and use $(i\omega)^2 = -\omega^2$ to obtain

$$\boxed{\nabla \times \big(\mu^{-1} \nabla \times \mathbf{E}\big) \;-\; \omega^2\, \varepsilon_c\, \mathbf{E} \;=\; \mathbf{0}}. \tag{15}$$

which is the Electric-field wave equation.

From equation (12), the electric field can be expressed as

$$\mathbf{E}(\mathbf{r},\omega) = \frac{i}{\omega}\, \varepsilon_c^{-1}(\mathbf{r},\omega)\, \big(\nabla \times \mathbf{H}(\mathbf{r},\omega)\big). \tag{16}$$

Insert into (9):

$$\nabla \times \left(\frac{i}{\omega} \varepsilon_c^{-1} \nabla \times \mathbf{H}\right) = i\omega\, \mu\, \mathbf{H}, \tag{17}$$

which yields

$$\boxed{\nabla \times \big(\varepsilon_c^{-1} \nabla \times \mathbf{H}\big) \;-\; \omega^2\, \mu\, \mathbf{H} \;=\; \mathbf{0}}. \tag{18}$$

which is the Magnetic-field wave equation.

Equations (15)–(18) are exact and fully vectorial in inhomogeneous media.

4 Reduction to a 2D Guided-Mode Eigenproblem

We consider a straight waveguide with material profiles $\varepsilon_c(x,y)$ and $\mu(x,y)$ that are invariant along z. In the frequency domain with $e^{-i\omega t}$ time dependence, guided modes are assumed to have the separable form

$$\mathbf{E}(x,y,z,t) = \mathbf{e}(x,y)\, e^{i\beta z} e^{-i\omega t}, \qquad \mathbf{H}(x,y,z,t) = \mathbf{h}(x,y)\, e^{i\beta z} e^{-i\omega t}, \tag{19}$$

where β is the propagation constant and $n_{\text{eff}} = \beta/k_0$ with $k_0 = \omega/c$.

Define the transverse gradient

$$\boxed{\nabla_t \equiv \hat{\mathbf{x}}\,\partial_x + \hat{\mathbf{y}}\,\partial_y}, \tag{20}$$

so that, using $\partial_z \to i\beta$,

$$\nabla = \nabla_t + \hat{\mathbf{z}}\,\partial_z = \nabla_t + i\beta\,\hat{\mathbf{z}}. \tag{21}$$

We also introduce the β-dependent curl operator

$$\boxed{\mathcal{C}_\beta[\cdot] \;\equiv\; (\nabla_t + i\beta\hat{\mathbf{z}}) \times (\cdot)}, \tag{22}$$

which acts only on (x,y)-dependent vector fields.

Starting from the frequency-domain electric curl–curl wave equation

$$\nabla \times \big(\mu^{-1} \nabla \times \mathbf{E}\big) - \omega^2\, \varepsilon_c\, \mathbf{E} = \mathbf{0}, \tag{23}$$

and substituting the modal ansatz (19), we first compute

$$\begin{aligned}
\nabla \times \mathbf{E} &= \big(\nabla_t + \hat{\mathbf{z}}\,\partial_z\big) \times \big(\mathbf{e}\, e^{i\beta z}\big) \\
&= \nabla_t \times (\mathbf{e}\, e^{i\beta z}) \;+\; \hat{\mathbf{z}} \times \partial_z(\mathbf{e}\, e^{i\beta z}) \\
&= e^{i\beta z}(\nabla_t \times \mathbf{e}) \;+\; e^{i\beta z}\, i\beta\, \hat{\mathbf{z}} \times \mathbf{e} \\
&= \big[(\nabla_t \times \mathbf{e}) + i\beta\,\hat{\mathbf{z}} \times \mathbf{e}\big]\, e^{i\beta z} \\
&= \mathcal{C}_\beta[\mathbf{e}]\, e^{i\beta z}.
\end{aligned} \tag{24}$$

Since $\mu^{-1} = \mu^{-1}(x,y)$ depends only on (x,y),

$$\mu^{-1}\, \nabla \times \mathbf{E} = \big(\mu^{-1}\, \mathcal{C}_\beta[\mathbf{e}]\big)\, e^{i\beta z}. \tag{25}$$

Let $\mathbf{Q}(x,y) \equiv \mu^{-1}(x,y)\, \mathcal{C}_\beta[\mathbf{e}(x,y)]$. Applying the curl again:

$$\begin{aligned}
\nabla \times \big(\mathbf{Q}\, e^{i\beta z}\big) &= (\nabla e^{i\beta z}) \times \mathbf{Q} + e^{i\beta z}\, \nabla \times \mathbf{Q} \\
&= i\beta\, e^{i\beta z}\, \hat{\mathbf{z}} \times \mathbf{Q} + e^{i\beta z}\, (\nabla_t \times \mathbf{Q}) \\
&= e^{i\beta z}\, (\nabla_t + i\beta\hat{\mathbf{z}}) \times \mathbf{Q} \\
&= e^{i\beta z}\, \mathcal{C}_\beta[\mathbf{Q}].
\end{aligned} \tag{26}$$

Restoring $\mathbf{Q} = \mu^{-1}\mathcal{C}_\beta[\mathbf{e}]$ gives

$$\nabla \times \big(\mu^{-1}\nabla \times \mathbf{E}\big) = e^{i\beta z}\, \mathcal{C}_\beta\big[\, \mu^{-1}\, \mathcal{C}_\beta[\mathbf{e}]\, \big]. \tag{27}$$

The second term in (23) is

$$\omega^2\, \varepsilon_c\, \mathbf{E} = \omega^2\, \varepsilon_c(x, y)\, \mathbf{e}(x, y)\, e^{i\beta z}. \tag{28}$$

Substituting (27) and (28) into (23) yields

$$e^{i\beta z}\, \big\{\mathcal{C}_\beta\big[\, \mu^{-1}\, \mathcal{C}_\beta[\mathbf{e}]\, \big] - \omega^2\, \varepsilon_c\, \mathbf{e}\big\} = \mathbf{0}. \tag{29}$$

Since $e^{i\beta z} \neq 0$, the bracketed term must vanish. With $k_0 = \omega/c$, the result is the purely 2D vector PDE

$$\boxed{\mathcal{C}_\beta\big[\, \mu^{-1}\, \mathcal{C}_\beta[\mathbf{e}]\, \big] - k_0^2\, \varepsilon_c\, \mathbf{e} = \mathbf{0}, \qquad k_0 = \omega/c.} \tag{30}$$

This is a *two-dimensional vectorial eigenproblem* in (x, y) for the modal profile $\mathbf{e}(x, y)$, with β entering as the eigenvalue parameter through $\mathcal{C}_\beta$.

5 Operator Form and the Final 2D Eigenproblem

We now rewrite the 2D modal equation from Sec. 4 in a compact operator form. The starting point is

$$\mathcal{C}_\beta\big[\mu^{-1}\, \mathcal{C}_\beta[\mathbf{e}]\big] - k_0^2\, \varepsilon_c\, \mathbf{e} = \mathbf{0}, \tag{31}$$

where the β–dependent curl operator acts on (x, y)–dependent fields as

$$\mathcal{C}_\beta[\cdot] \equiv \big(\nabla_t + i\beta\, \hat{\mathbf{z}}\big) \times (\cdot). \tag{32}$$

By bilinearity of the curl and invariance of $\mu^{-1}(x, y)$ along z we have

$$\mathcal{C}_\beta[\mathbf{v}] = \nabla_t \times \mathbf{v} + i\beta\, \hat{\mathbf{z}} \times \mathbf{v}. \tag{33}$$

Applying (33) to $\mathbf{e}$ and then again to $\mu^{-1}\mathcal{C}_\beta[\mathbf{e}]$ gives

$$\begin{aligned}\mathcal{C}_\beta\big[\mu^{-1}\, \mathcal{C}_\beta[\mathbf{e}]\big] &= \nabla_t \times \big(\mu^{-1}\, \nabla_t \times \mathbf{e}\big) + i\beta\, \nabla_t \times \big(\mu^{-1}\, \hat{\mathbf{z}} \times \mathbf{e}\big) \\ &\quad + i\beta\, \hat{\mathbf{z}} \times \big(\mu^{-1}\, \nabla_t \times \mathbf{e}\big) - \beta^2\, \hat{\mathbf{z}} \times \big(\mu^{-1}\, \hat{\mathbf{z}} \times \mathbf{e}\big).\end{aligned} \tag{34}$$

Substituting (34) into (31) and grouping by powers of β yields

$$\big[\nabla_t \times \mu^{-1}\, \nabla_t \times - k_0^2\, \varepsilon_c\, \mathbf{I}\big]\mathbf{e} + i\beta\big[\nabla_t \times \mu^{-1}(\hat{\mathbf{z}}\times) + (\hat{\mathbf{z}}\times)\, \mu^{-1}\, \nabla_t \times\big]\mathbf{e} - \beta^2(\hat{\mathbf{z}}\times)\, \mu^{-1}\, (\hat{\mathbf{z}}\times)\mathbf{e} = \mathbf{0}. \tag{35}$$

This has the compact quadratic eigenproblem form

$$\boxed{\big(-\beta^2\, \mathbf{W} + i\beta\, \mathbf{C} + \mathbf{K}\big)\, \mathbf{e} = \mathbf{0},} \tag{36}$$

with

$$\boxed{\begin{aligned}\mathbf{W} &= (\hat{\mathbf{z}}\times)\,\mu^{-1}\,(\hat{\mathbf{z}}\times),\\ \mathbf{C} &= \nabla_t \times \mu^{-1}(\hat{\mathbf{z}}\times) + (\hat{\mathbf{z}}\times)\,\mu^{-1}\,\nabla_t\times,\\ \mathbf{K} &= \nabla_t \times \mu^{-1}\,\nabla_t \times - k_0^2\,\varepsilon_c\,\mathbf{I},\end{aligned}} \tag{37}$$

and, equivalently,

$$\boxed{\mathbf{L}(\varepsilon_c, \mu; \omega, \beta) \equiv \mathbf{K} + i\beta\,\mathbf{C} \quad \Rightarrow \quad \mathbf{L}\,\mathbf{e} = \beta^2\,\mathbf{W}\,\mathbf{e},} \tag{38}$$

which is a generalized eigenvalue problem in β^2. The effective index follows as $n_{\text{eff}} = \beta/k_0$. The $\hat{\mathbf{z}}\times$ operator is represented by

$$(\hat{\mathbf{z}}\times) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad (\hat{\mathbf{z}}\times)\,\mathbf{v} = \begin{bmatrix} -v_y \\ v_x \\ 0 \end{bmatrix}, \quad (\hat{\mathbf{z}}\times)^2 = -\,\Pi_t, \tag{39}$$

where $\Pi_t = \mathrm{diag}(1, 1, 0)$ projects onto the transverse (x, y) subspace. If μ is scalar, isotropic, and real, then $\mu^{-1} = \mu_0^{-1}\mathbf{I}$ and

$$\mathbf{W} = \mu_0^{-1}(\hat{\mathbf{z}}\times)(\hat{\mathbf{z}}\times), \quad \mathbf{C} = \mu_0^{-1}\big(\nabla_t\times(\hat{\mathbf{z}}\times)+(\hat{\mathbf{z}}\times)\nabla_t\times\big), \quad \mathbf{K} = \mu_0^{-1}\,\nabla_t\times\nabla_t\times-k_0^2\,\varepsilon_c\,\mathbf{I}, \tag{40}$$

so that for real, positive ε_c and suitable boundaries, $\mathbf{K}$ is Hermitian and $\mathbf{W}$ is positive semidefinite.

6 Numerical Method (discretization and solver)

We discretize the transverse domain on a uniform, cell–centered Cartesian grid with spacing $\Delta x = \Delta y =$ `DX` and a rectangular computational window of half–widths `PAD` around the core. Dirichlet walls (fields set to zero) are placed at the box boundary; to suppress boundary interaction we use `PAD` $\geq 5\,\mu\text{m}$.

Spatial derivatives use first–order backward differences D_x, D_y lifted to 2D by Kronecker products. With isotropic $\mu = \mu_r\mu_0$ the discrete full–vector transverse formulation for the unknowns $\{E_x, E_y\}$ reads

$$\mathbf{M}\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \beta^2 \begin{bmatrix} E_x \\ E_y \end{bmatrix}, \quad \mathbf{M} = \begin{bmatrix} -D_y^\top \mu^{-1} D_y + k_0^2\varepsilon & D_y^\top \mu^{-1} D_x \\ D_x^\top \mu^{-1} D_y & -D_x^\top \mu^{-1} D_x + k_0^2\varepsilon \end{bmatrix}.$$

We solve this Hermitian EVP with ARPACK (`eigsh`) in shift–invert mode, targeting $\sigma = (\tilde{n}\,k_0)^2$ with $\tilde{n} \approx$ the expected n_{eff} of the fundamental TE mode. The longitudinal field E_z and the magnetic field (H_x, H_y, H_z) are then reconstructed from (E_x, E_y) via Gauss' law and Faraday's law.

For validation and in practical design I would normally use a standard mode solver (such as EMpy, MEEP/MPB, or a commercial foundry PDK tool). However, in this work I implemented a full–vector FDFD eigenmode solver from scratch in Python, in order to demonstrate my understanding of the underlying formulation (Maxwell curl–curl equation, discretization, and sparse eigenvalue problem).

7 Complex Poynting Vector, Normalization, and Final Eigenproblem

The complex Poynting vector is defined as

$$\mathbf{S}(\mathbf{r}) = \frac{1}{2}\,\mathbf{E}(\mathbf{r}) \times \mathbf{H}^*(\mathbf{r}) \tag{41}$$

whose real part $\Re\{\mathbf{S}\}$ represents the time-averaged power flux density in W/m^2, and whose imaginary part $\Im\{\mathbf{S}\}$ corresponds to reactive power associated with oscillatory stored energy. For a guided mode, the total forward-propagating power is obtained from the z–component of the time-averaged Poynting vector as

$$P = \iint_{R^2} \Re\{S_z(x,y)\}\,dx\,dy, \tag{42}$$

and in lossless, source-free guides, mode orthogonality and normalization conditions are naturally expressed through the $\Re\{S_z\}$ flux.

With $\varepsilon_c(x,y)$ and $\mu(x,y)$ specified, the 2D modal equation

$$\mathcal{C}_\beta\big[\mu^{-1}\,\mathcal{C}_\beta[\mathbf{e}]\big] - k_0^2\,\varepsilon_c\,\mathbf{e} = \mathbf{0}, \tag{43}$$

follows directly from Maxwell's curl–curl formulation under the $e^{i\beta z}$ ansatz.

Starting from

$$\mathcal{C}_\beta\big[\mu^{-1}\,\mathcal{C}_\beta[\mathbf{e}]\big] - k_0^2\,\varepsilon_c\,\mathbf{e} = \mathbf{0}, \qquad \mathcal{C}_\beta[\mathbf{v}] \equiv \nabla_t \times \mathbf{v} + i\beta\,\hat{\mathbf{z}} \times \mathbf{v}, \tag{44}$$

set

$$\mathbf{u} \;\equiv\; \mu^{-1}\,\mathcal{C}_\beta[\mathbf{e}] \;=\; \mu^{-1}\,(\nabla_t \times \mathbf{e}) \;+\; i\beta\,\mu^{-1}\,(\hat{\mathbf{z}} \times \mathbf{e}). \tag{45}$$

Apply $\mathcal{C}_\beta$ to $\mathbf{u}$:

$$\begin{aligned}
\mathcal{C}_\beta[\mathbf{u}] &= \nabla_t \times \mathbf{u} + i\beta\,\hat{\mathbf{z}} \times \mathbf{u} \\
&= \nabla_t \times \big(\mu^{-1}\,\nabla_t \times \mathbf{e}\big) \;+\; i\beta\,\nabla_t \times \big(\mu^{-1}\,\hat{\mathbf{z}} \times \mathbf{e}\big) \\
&\quad +\; i\beta\,\hat{\mathbf{z}} \times \big(\mu^{-1}\,\nabla_t \times \mathbf{e}\big) \;+\; (i\beta)^2\,\hat{\mathbf{z}} \times \big(\mu^{-1}\,\hat{\mathbf{z}} \times \mathbf{e}\big) \\
&= \nabla_t \times \big(\mu^{-1}\,\nabla_t \times \mathbf{e}\big) \;+\; i\beta\Big[\nabla_t \times \big(\mu^{-1}\,\hat{\mathbf{z}} \times \mathbf{e}\big) \;+\; \hat{\mathbf{z}} \times \big(\mu^{-1}\,\nabla_t \times \mathbf{e}\big)\Big] \;-\; \beta^2\,\hat{\mathbf{z}} \times \big(\mu^{-1}\,\hat{\mathbf{z}} \times \mathbf{e}\big).
\end{aligned} \tag{46}$$

Insert (46) into (44) and move the mass term to the left:

$$\Big[\nabla_t \times \mu^{-1}\,\nabla_t \times \;-\; k_0^2\,\varepsilon_c\,\mathbf{I}\Big]\mathbf{e} \;+\; i\beta\Big[\nabla_t \times \mu^{-1}(\hat{\mathbf{z}}\times) \;+\; (\hat{\mathbf{z}}\times)\mu^{-1}\,\nabla_t \times\Big]\mathbf{e} \;-\; \beta^2\,(\hat{\mathbf{z}}\times)\mu^{-1}(\hat{\mathbf{z}}\times)\,\mathbf{e} \;=\; \mathbf{0}. \tag{47}$$

Define the operators

$$\mathbf{W} \;\equiv\; (\hat{\mathbf{z}}\times)\,\mu^{-1}\,(\hat{\mathbf{z}}\times), \qquad \mathbf{C} \;\equiv\; \nabla_t\times\mu^{-1}(\hat{\mathbf{z}}\times) + (\hat{\mathbf{z}}\times)\,\mu^{-1}\,\nabla_t\times, \qquad \mathbf{K} \;\equiv\; \nabla_t\times\mu^{-1}\,\nabla_t\times - k_0^2\,\varepsilon_c\,\mathbf{I}, \tag{48}$$

so that (47) becomes

$$\boxed{\big(-\beta^2\,\mathbf{W} + i\beta\,\mathbf{C} + \mathbf{K}\big)\,\mathbf{e} \;=\; \mathbf{0}.} \tag{49}$$

where $\mathbf{W}$, $\mathbf{C}$, and $\mathbf{K}$ are given in Eqs. (37)–(38). Equation (49) is exact for any material distribution $\varepsilon_c(x,y)$, $\mu(x,y)$ and any open or closed transverse boundary conditions.

In a dielectric waveguide with (x,y) cross section $\Omega \subset R^2$ and material maps $\varepsilon_c(x,y)$ and $\mu(x,y)$ that are invariant along z, the modal electric and magnetic fields are assumed to have the separable form

$$\mathbf{E}(x,y,z) = \mathbf{e}(x,y)\,e^{i\beta z}, \qquad \mathbf{H}(x,y,z) = \mathbf{h}(x,y)\,e^{i\beta z}. \tag{50}$$

The frequency-domain Maxwell equation for the electric field is

$$\nabla \times \mu^{-1}\nabla \times \mathbf{E} - \omega^2 \varepsilon_c\,\mathbf{E} = \mathbf{0}. \tag{51}$$

Substituting (50) into (51) and using $\partial_z \to i\beta$ gives the purely transverse equation

$$\mathcal{C}_\beta\big[\mu^{-1}\,\mathcal{C}_\beta[\mathbf{e}]\big] - k_0^2\,\varepsilon_c\,\mathbf{e} = \mathbf{0}, \tag{52}$$

where

$$\mathcal{C}_\beta[\cdot] \equiv (\nabla_t + i\beta\hat{\mathbf{z}}) \times (\cdot), \qquad \nabla_t \equiv \hat{\mathbf{x}}\,\partial_x + \hat{\mathbf{y}}\,\partial_y, \qquad k_0 = \frac{\omega}{c}. \tag{53}$$

By bilinearity of the curl operator,

$$\mathcal{C}_\beta[\mathbf{v}] = \nabla_t \times \mathbf{v} + i\beta\,\hat{\mathbf{z}} \times \mathbf{v}. \tag{54}$$

Let

$$\mathbf{u} \equiv \mu^{-1}\,\mathcal{C}_\beta[\mathbf{e}] = \mu^{-1}\,(\nabla_t \times \mathbf{e}) + i\beta\,\mu^{-1}\,(\hat{\mathbf{z}} \times \mathbf{e}). \tag{55}$$

Applying $\mathcal{C}_\beta$ to $\mathbf{u}$ yields

$$\begin{aligned}
\mathcal{C}_\beta[\mathbf{u}] &= \nabla_t \times \mathbf{u} + i\beta\,\hat{\mathbf{z}} \times \mathbf{u} && (56)\\
&= \nabla_t \times \big(\mu^{-1}\,\nabla_t \times \mathbf{e}\big) + i\beta\,\nabla_t \times \big(\mu^{-1}\,\hat{\mathbf{z}} \times \mathbf{e}\big) \\
&\quad + i\beta\,\hat{\mathbf{z}} \times \big(\mu^{-1}\,\nabla_t \times \mathbf{e}\big) + (i\beta)^2\,\hat{\mathbf{z}} \times \big(\mu^{-1}\,\hat{\mathbf{z}} \times \mathbf{e}\big) && (57)\\
&= \nabla_t \times \big(\mu^{-1}\,\nabla_t \times \mathbf{e}\big) + i\beta\Big[\nabla_t \times \big(\mu^{-1}\,\hat{\mathbf{z}} \times \mathbf{e}\big) + \hat{\mathbf{z}} \times \big(\mu^{-1}\,\nabla_t \times \mathbf{e}\big)\Big] - \beta^2\,\hat{\mathbf{z}} \times \big(\mu^{-1}\,\hat{\mathbf{z}} \times \mathbf{e}\big). && (58)
\end{aligned}$$

Substituting (58) into (52) and moving the mass term to the left gives

$$\begin{aligned}
&\big[\nabla_t \times \mu^{-1}\,\nabla_t \times - k_0^2\,\varepsilon_c\,\mathbf{I}\big]\,\mathbf{e} + i\beta\big[\nabla_t \times \mu^{-1}(\hat{\mathbf{z}}\times) + (\hat{\mathbf{z}}\times)\mu^{-1}\,\nabla_t \times\,\big]\,\mathbf{e} \\
&\quad - \beta^2\,(\hat{\mathbf{z}}\times)\,\mu^{-1}\,(\hat{\mathbf{z}}\times)\,\mathbf{e} = \mathbf{0}.
\end{aligned} \tag{59}$$

We now define the operators

$$\mathbf{W} \equiv (\hat{\mathbf{z}}\times)\,\mu^{-1}\,(\hat{\mathbf{z}}\times), \qquad \mathbf{C} \equiv \nabla_t\times\mu^{-1}(\hat{\mathbf{z}}\times) + (\hat{\mathbf{z}}\times)\,\mu^{-1}\,\nabla_t\times, \qquad \mathbf{K} \equiv \nabla_t\times\mu^{-1}\,\nabla_t\times - k_0^2\,\varepsilon_c\,\mathbf{I}. \tag{60}$$

In terms of these, (59) becomes the quadratic eigenvalue problem

$$-\beta^2\,\mathbf{W}\,\mathbf{e} + i\beta\,\mathbf{C}\,\mathbf{e} + \mathbf{K}\,\mathbf{e} = \mathbf{0}. \tag{61}$$

This is equivalently written as the generalized eigenvalue problem

$$\mathbf{L}(\varepsilon_c, \mu; \omega, \beta)\,\mathbf{e} = \beta^2\,\mathbf{W}\,\mathbf{e}, \qquad \mathbf{L} \equiv \mathbf{K} + i\beta\,\mathbf{C}. \tag{62}$$

The effective index is defined by

$$n_{\text{eff}} = \frac{\beta}{k_0}. \tag{63}$$

In the presence of perfectly matched layer boundaries, the material parameters ε_c and/or μ become complex-valued in the outer regions, but the operator form (61) remains valid. The magnetic field profile follows from

$$\mathbf{h} = \frac{1}{i\omega}\,\mu^{-1}\,(\nabla_t + i\beta\,\hat{\mathbf{z}}) \times \mathbf{e}. \tag{64}$$

The complex Poynting vector is

$$\mathbf{S} = \frac{1}{2}\,\mathbf{e} \times \mathbf{h}^*, \tag{65}$$

and the forward power used for normalization is

$$P = \frac{1}{2}\iint_{\Omega} \Re\{\mathbf{e} \times \mathbf{h}^* \cdot \hat{\mathbf{z}}\}\,dx\,dy. \tag{66}$$

Thus, the final continuous problem reads

$$\boxed{\Big(-\beta^2\,\mathbf{W}(\mu) + i\beta\,\mathbf{C}(\mu, \nabla_t) + \mathbf{K}(\varepsilon_c, \mu, \nabla_t; \omega)\Big)\,\mathbf{e}(x, y) = \mathbf{0},} \qquad n_{\text{eff}} = \frac{\beta}{k_0}. \tag{67}$$

Numerical implementation used in this work (consistency with code). In the solver we discretize on a uniform, cell–centered Cartesian grid with spacing $\Delta x = \Delta y$ and apply Dirichlet walls at the padded box boundary. After eliminating E_z via Gauss' law and $\mathbf{H}$ via Faraday's law, the full–vector transverse–$\mathbf{E}$ formulation yields a *linear* eigenproblem for the unknown interior fields $\{E_x, E_y\}$:

$$\underbrace{\begin{bmatrix} -D_y^{\top}\,\mu^{-1} D_y + k_0^2\,\varepsilon & D_y^{\top}\,\mu^{-1} D_x \\ D_x^{\top}\,\mu^{-1} D_y & -D_x^{\top}\,\mu^{-1} D_x + k_0^2\,\varepsilon \end{bmatrix}}_{\mathbf{M}} \begin{bmatrix} E_x \\ E_y \end{bmatrix} = \beta^2 \begin{bmatrix} E_x \\ E_y \end{bmatrix}. \tag{68}$$

Here D_x, D_y are the discrete first–difference operators (lifted to 2D by Kronecker products), and $\varepsilon(x, y)$, $\mu(x, y)$ are sampled on the grid. We solve (68) with ARPACK in shift–invert mode, targeting $\sigma = (\tilde{n}\,k_0)^2$. The longitudinal field E_z and magnetic components (H_x, H_y, H_z) are reconstructed *a posteriori* from (E_x, E_y) using Gauss' law and Faraday's law. This discrete transverse–$\mathbf{E}$ EVP is algebraically equivalent to the quadratic operator form (67) under the standard elimination, and therefore yields the same guided-mode effective indices and vectorial profiles.

8 Geometry and Materials

We model a rectangular silicon (Si) core of width $w = 450\,\mathrm{nm}$ and thickness $t = 220\,\mathrm{nm}$, embedded in silica (SiO_2) on all sides. Material models:

- SiO_2: 3–term Sellmeier (Malitson), evaluated at the requested wavelength.
- Si: near–IR real index interpolated from anchor points at 1.30–1.60 µm.

We use `DX` $= 20\,\mathrm{nm}$, `PAD` $= 5\,\mu\mathrm{m}$, and compute k eigenpairs per wavelength with `KMODES=12`.

9 Results

Mode labeling. TE_0 is the highest–n_{eff} mode whose energy in the core is predominantly in E_x (TE score ≥ 0.55). TM_0 is the highest–n_{eff} among the remaining modes with predominant E_y (TM score ≥ 0.55). We also report the fraction of total electromagnetic energy in E_z. Small values indicate weakly longitudinal fields. As a physics check, we measure the propagation self–overlap in a straight and uniform guide. A true eigenmode keeps overlap ≈ 1 along z.

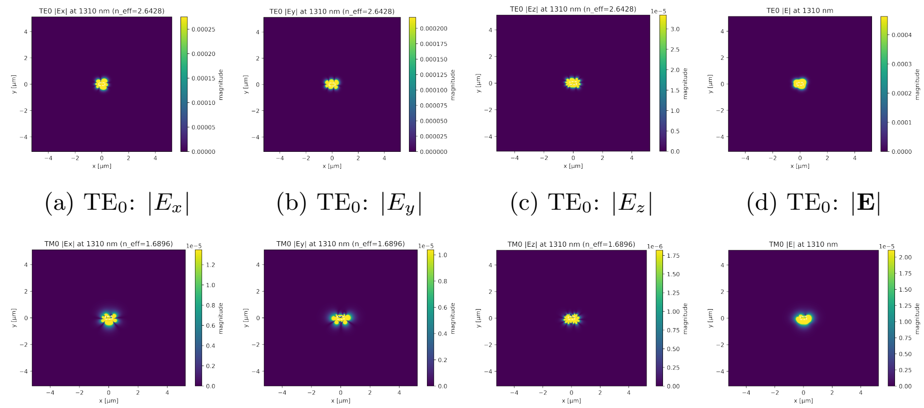
Metrics. TE/TM labeling uses core–integrated transverse components

$$\mathrm{TE\_score} = \frac{\int_{\mathrm{core}} |E_x|^2\,dx\,dy}{\int_{\mathrm{core}} (|E_x|^2 + |E_y|^2)\,dx\,dy}, \qquad \mathrm{TM\_score} = 1 - \mathrm{TE\_score}.$$

Propagation self–overlap is the normalized ε_r–weighted inner product

$$\mathcal{O}(z) = \frac{\left| \iint \varepsilon_r\,\mathbf{E}(x,y,0)^* \cdot \mathbf{E}(x,y,z)\,dx\,dy \right|}{\sqrt{\iint \varepsilon_r |\mathbf{E}(x,y,0)|^2\,dx\,dy}\,\sqrt{\iint \varepsilon_r |\mathbf{E}(x,y,z)|^2\,dx\,dy}},$$

which stays ≈ 1 for a true eigenmode in a uniform guide.



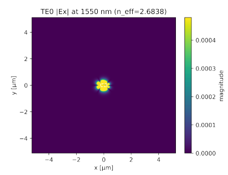
(a) TE_0: $|E_x|$ (b) TE_0: $|E_y|$ (c) TE_0: $|E_z|$ (d) TE_0: $|\mathbf{E}|$

(e) TM_0: $|E_x|$ (f) TM_0: $|E_y|$ (g) TM_0: $|E_z|$ (h) TM_0: $|\mathbf{E}|$

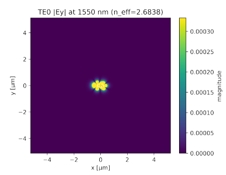
Figure 1: Vectorial mode profiles at 1310 nm. Colors show field magnitude using the viridis colormap. Yellow marks the highest magnitude within each panel and dark purple the lowest. Each panel is normalized to its own maximum, so colors are not comparable across panels. Top row (TE_0) has $n_{\mathrm{eff}} = 2.6428$, TE score 0.662, and E_z fraction 0.034. Bottom row (TM_0) has $n_{\mathrm{eff}} = 1.6896$, TM score 0.854, and E_z fraction 0.032.

1310 nm (summary). TE_0 has $n_{\mathrm{eff}} = 2.6428$, TE score 0.662, and E_z fraction 0.034. Residuals were curl 4.40×10^{-9} and Gauss 6.65×10^{-10}. TM_0 has $n_{\mathrm{eff}} = 1.6896$, TM score 0.854, and E_z fraction 0.032. Residuals were curl 4.18×10^{-9} and Gauss 2.83×10^{-10}.

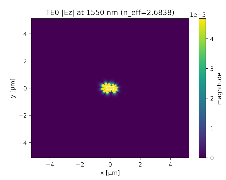
9.1 Vectorial field profiles at 1550 nm



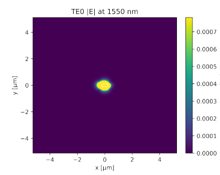
(a) TE_0: $|E_x|$



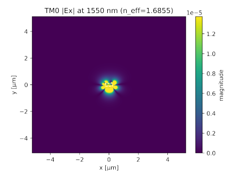
(b) TE_0: $|E_y|$



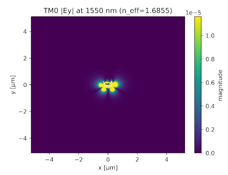
(c) TE_0: $|E_z|$



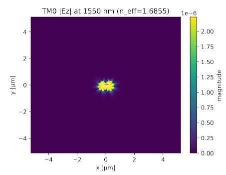
(d) TE_0: $|\mathbf{E}|$



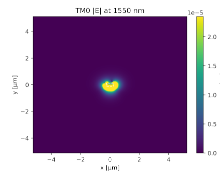
(e) TM_0: $|E_x|$



(f) TM_0: $|E_y|$



(g) TM_0: $|E_z|$

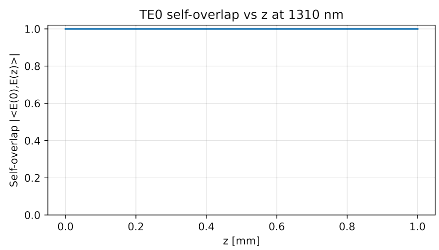


(h) TM_0: $|\mathbf{E}|$

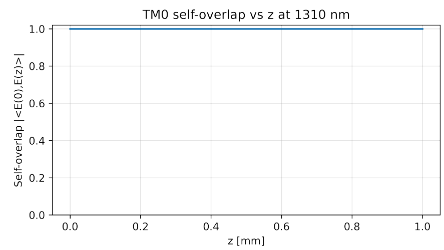
Figure 2: Vectorial mode profiles at 1550 nm. Colors show field magnitude using the viridis colormap. Yellow marks the highest magnitude within each panel and dark purple the lowest. Each panel is normalized to its own maximum, so colors are not comparable across panels. Top row (TE_0) has $n_{\mathrm{eff}} = 2.6838$, TE score 0.603, and E_z fraction 0.028. Bottom row (TM_0) has $n_{\mathrm{eff}} = 1.6855$, TM score 0.854, and E_z fraction 0.023.

1550 nm (summary). TE_0 has $n_{\mathrm{eff}} = 2.6838$, TE score 0.603, and E_z fraction 0.028. Residuals were curl 2.80×10^{-9} and Gauss 7.90×10^{-10}. TM_0 has $n_{\mathrm{eff}} = 1.6855$, TM score 0.854, and E_z fraction 0.023. Residuals were curl 1.73×10^{-9} and Gauss 4.51×10^{-10}.

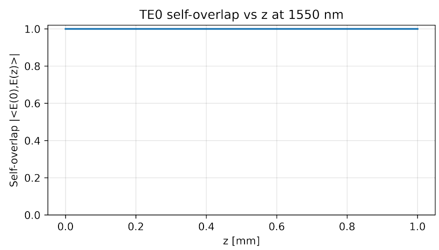
We propagated each mode in the same straight and uniform waveguide and measured the normalized self–overlap $\mathcal{O}(z) = |\langle \mathbf{f}(0), \mathbf{f}(z) \rangle| / (\|\mathbf{f}(0)\| \, \|\mathbf{f}(z)\|)$ along z. For an eigenmode $\mathcal{O}(z)$ remains close to one. A non–eigen launch would show beating.



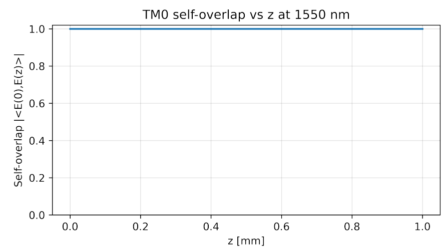
(a) TE_0 self–overlap at 1310 nm. The curve is flat at approximately one.



(b) TM_0 self–overlap at 1310 nm. The curve is flat at approximately one.



(c) TE_0 self–overlap at 1550 nm. The curve is flat at approximately one.



(d) TM_0 self–overlap at 1550 nm. The curve is flat at approximately one.

Figure 3: Propagation invariance. The launched TE_0 and TM_0 fields preserve their transverse profile in a straight and uniform guide.

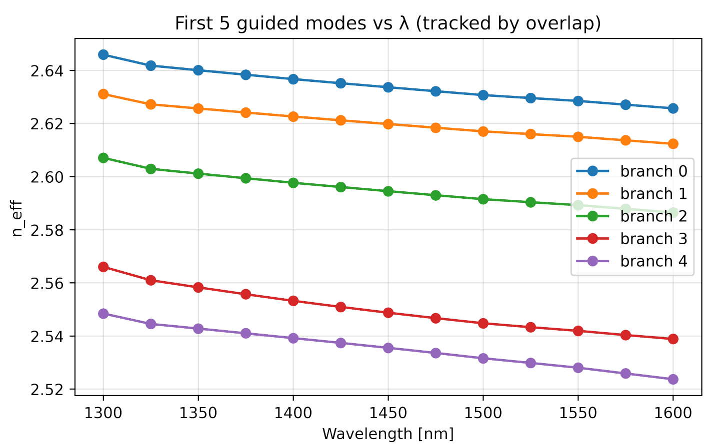


Figure 4: Tracked dispersion of the five highest-index guided branches between 1300 and 1600 nm. Curves are tracked by field-overlap continuity across wavelength so each color follows the same physical branch. Colors identify branches only. Blue is branch 0. Orange is branch 1. Green is branch 2. Red is branch 3. Purple is branch 4.

Figure 4 plots the effective indices of the five highest-index guided branches from 1300 to 1600 nm, with branch continuity enforced by field–overlap tracking. All curves decrease gently with wavelength, consistent with normal modal dispersion in high–contrast Si/SiO_2 waveguides. The top curve (branch 0) is the fundamental TE-like mode and stays in the mid–2.5 range across this window (about 2.63 at 1300 nm to 2.57 at 1600 nm). Branches 1–4 are higher–order TE-like modes that also soften with wavelength and move toward cutoff. A point that falls to the SiO_2 cladding index ($n_{\mathrm{clad}} \approx 1.44$) is no longer guided.

Colors (blue, orange, green, red, purple) label branches. The TM_0 branch has a much lower index in this geometry ($n_{\mathrm{eff}} \approx 1.68$–$1.70$ over 1310–1550 nm) and is therefore *not* among the five highest-index branches shown here.

10 Discussion: why 450 nm width?

The canonical 220 nm × 450 nm silicon core (fully clad in SiO_2) strikes a practical balance between modal control, loss, and manufacturability. Around the O- and C-bands (1310 nm to 1550 nm), a 450 nm width keeps TE_0 strongly confined while pushing higher-order TE modes close to cutoff; TM_0 is weakly guided or near cutoff in many stacks. This delivers effectively single-mode, TE-preferred operation across the telecom window and simplifies polarization handling in large photonic circuits.

The same cross section supports tight routing with low bend radiation: confinement is strong enough that micron-scale bend radii incur modest excess loss, yet not so extreme that sidewall scattering dominates. From a fabrication standpoint, 450 nm offers a forgiving process window—typical linewidth and thickness variations (tens of nanometers) shift n_{eff} only moderately and do not abruptly trigger higher-order mode guidance.

Equally important is ecosystem maturity. Foundry PDKs, standard cell libraries (MMIs, crossings, filters), and grating-coupler designs have been extensively optimized around 450 nm, enabling predictable tape-outs, good yield, and interoperability. In short, 450 nm became an "industry standard" because it simultaneously provides robust single-TE operation over 1310 nm to 1550 nm, compact low-loss routing, tolerance to fabrication variations, and compatibility with widely adopted building blocks and design kits.